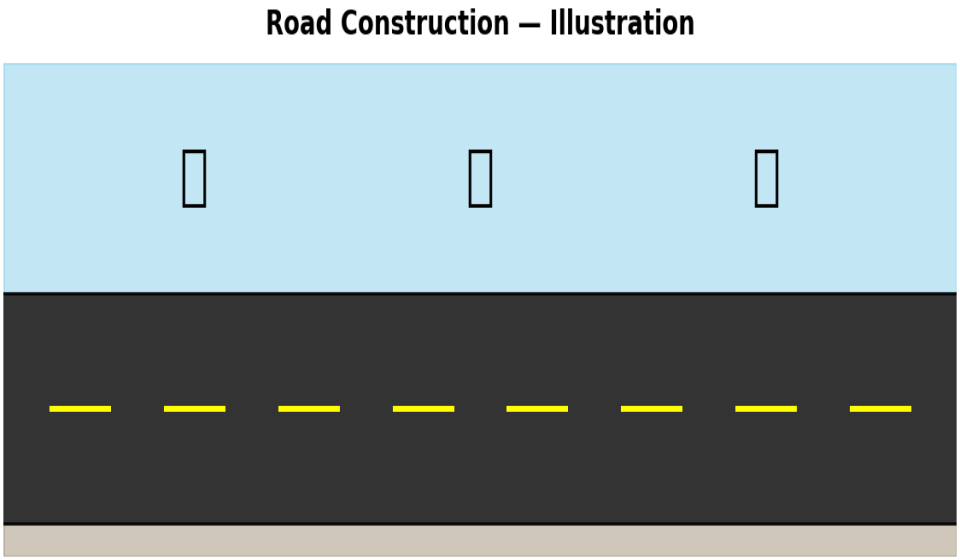
INDIAN INSTITUTE OF TECHNOLOGY (BHU) VARANASI
Department of Civil Engineering

AI-Driven Scheduling and Resource Allocation in Road Construction Projects



Conceptual Illustration — National Highway Road Construction Project

Subject: CE-401 — Construction Economics & Management

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Group Project — 18 Members

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1. Introduction & Motivation

Road construction is one of the most resource-intensive sectors in civil engineering. India's ambitious highway expansion programmes — such as Bharatmala Pariyojana — demand efficient scheduling and resource allocation to meet tight deadlines while controlling costs. Traditional approaches rely on manual planning using bar charts or rudimentary CPM, which fail to account for dynamic resource constraints, weather disruptions, and inter-task dependencies at scale.

Recent advances in Artificial Intelligence (AI) and Operations Research (OR) offer powerful alternatives. Constraint-Programming solvers can optimally allocate limited resources across dozens of interdependent tasks, while Machine Learning models can predict delays before they occur, enabling proactive risk management.

This project, RoadOpt AI, presents an integrated AI-driven system that combines Critical Path Method (CPM), Google OR-Tools CP-SAT solver for resource-constrained scheduling, and a Gradient Boosting Machine Learning model for delay-risk prediction — all wrapped in an interactive Streamlit dashboard for real-time what-if analysis.

2. Literature Review

Critical Path Method (CPM): Introduced by Kelley & Walker (1959), CPM remains the foundation of project scheduling in construction. It identifies the longest sequence of dependent tasks (the critical path) that determines the minimum project duration. However, classic CPM assumes unlimited resources, which is unrealistic in practice (Hegazy, 1999).

Resource-Constrained Project Scheduling (RCPSP): RCPSP extends CPM by imposing capacity limits on resources. It is NP-hard in general (Brucker et al., 1999). Modern solvers like Google OR-Tools employ Constraint Programming with SAT-backed search to find near-optimal solutions within seconds for problems with hundreds of tasks (Laborie et al., 2018).

AI in Construction Scheduling: Machine Learning has been applied to predict construction delays using historical data. Gondia et al. (2020) used Random Forests on 1,400 road projects and achieved 78% accuracy. Gradient Boosting methods (Chen & Guestrin, 2016) have shown superior performance for tabular prediction tasks, making them ideal for delay-risk estimation from structured project features.

Integrated Decision Support: Recent works (Faghihi et al., 2015; Amer & Golparvar-Fard, 2021) advocate integrating optimization with predictive analytics into interactive dashboards for construction managers, enabling scenario-based planning.

3. Problem Statement & Objectives

Problem: Given a road construction project with N interdependent tasks, each requiring specific resources (labor crews, excavators, pavers, etc.) with limited availability, determine an optimal weekly schedule that:

(a) Respects all task-dependency (precedence) constraints
(b) Does not exceed available resource capacities in any week
(c) Minimizes project completion time (makespan) and/or total cost
(d) Identifies and mitigates delay risks using AI prediction

3.1 Objectives

1. Implement Critical Path Method to establish a baseline schedule and identify critical tasks.
2. Solve the RCPSP using Google OR-Tools CP-SAT solver with configurable objectives (minimize time, minimize cost, balanced).
3. Train a Gradient Boosting ML model on synthetic historical data to predict task-level delay risk.
4. Build an interactive dashboard with Gantt charts, resource heatmaps, cost breakdowns, and what-if simulation capabilities.
5. Enable construction managers to make data-driven decisions under uncertainty.

3.2 Scope

The system models a realistic 24-task national highway construction project spanning survey, earthwork, pavement, structures, and finishing phases. Nine resource types are modelled with configurable capacities. The planning horizon is weekly over approximately one year.

3.3 Project Summary (Computed)

Metric	Value
Total Tasks	24
CPM Makespan (Ideal)	42 weeks
RCPSP Makespan (Resource-Constrained)	49 weeks
Solver Status	OPTIMAL
Total Estimated Cost	Rs.37,440,000
AI Model MAE	0.476 weeks
AI Model R^2	0.477
High-Risk Tasks	17

4. System Architecture & Methodology

RoadOpt AI follows a modular architecture with four core engines connected through a central configuration layer:

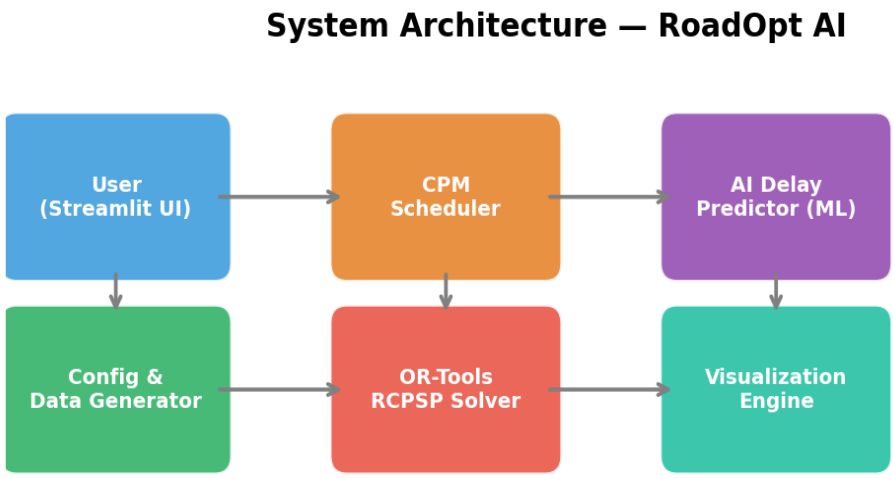

Figure 1: System Architecture of RoadOpt AI

4.1 Technology Stack

Component	Technology	Purpose
Optimization	Google OR-Tools CP-SAT	Resource-constrained scheduling
AI/ML	scikit-learn (GBR)	Delay-risk prediction
Dashboard	Streamlit + Plotly	Interactive web UI
Data Layer	Pandas + NumPy	Data generation & manipulation
Visualization	Plotly + Matplotlib	Charts & heatmaps
Language	Python 3.11	Core implementation

4.2 Methodology Pipeline

Step 1 — Data Generation: Synthetic task data with realistic durations, dependencies, and resource requirements modelled after Indian highway construction phases.
Step 2 — CPM Analysis: Forward and backward pass to compute earliest/latest starts and identify the critical path.
Step 3 — RCPSP Optimization: CP-SAT solver minimizes makespan subject to cumulative resource constraints and precedence relations.
Step 4 — AI Prediction: Gradient Boosting model trained on 3,000 synthetic historical records predicts per-task delay risk using 7 features.
Step 5 — Dashboard: Interactive Streamlit app presents results with what-if simulation.

5. Data Generation & Task Modeling

The project models a realistic National Highway construction project with 24 tasks organized across 6 phases: (1) Survey & Planning, (2) Site Preparation, (3) Earthwork, (4) Pavement Layers, (5) Structures, and (6) Finishing. Each task has a base duration (with ±20% random noise), a complexity rating (1–5), resource requirements, and predecessor dependencies.

5.1 Task List (Subset)

ID	Task Name	Duration (weeks)	Complexity	Predecessors
0	Site Survey & Soil Testing	3	2	—
1	Environmental Impact Assessm	3	3	0
2	Design & Engineering Approva	3	4	0, 1
3	Land Acquisition & Clearance	5	5	2
4	Site Clearing & Grubbing	3	2	3
5	Demolition of Existing Struc	2	3	3
6	Temporary Drainage Setup	2	2	3
7	Utility Relocation	3	4	3
8	Embankment Construction	5	3	4, 5, 6
9	Cut & Fill Operations	3	3	4, 5
10	Subgrade Preparation	3	3	8, 9
11	Granular Sub-Base (GSB) Laye	3	3	10

Table 1: First 12 of 24 project tasks (remaining tasks follow similar structure)

5.2 Resource Pool

Resource	Available Units	Cost/Week (Rs.)
Labor Crew	8	Rs.50,000
Excavator	4	Rs.120,000
Bulldozer	3	Rs.100,000
Asphalt Paver	2	Rs.150,000
Roller Compactor	3	Rs.80,000
Dump Truck	6	Rs.40,000
Concrete Mixer	3	Rs.60,000
Crane	2	Rs.200,000
Surveyor Team	2	Rs.70,000

Table 2: Resource pool with capacities and weekly costs

5.3 Historical Data for ML

A synthetic dataset of 2,000 historical task records was generated for training the delay-prediction model. Each record contains 7 features (task complexity, resource utilization, weather risk, dependency depth, critical-path membership, crew experience, and material availability) and a target variable (delay in weeks, 0–6).

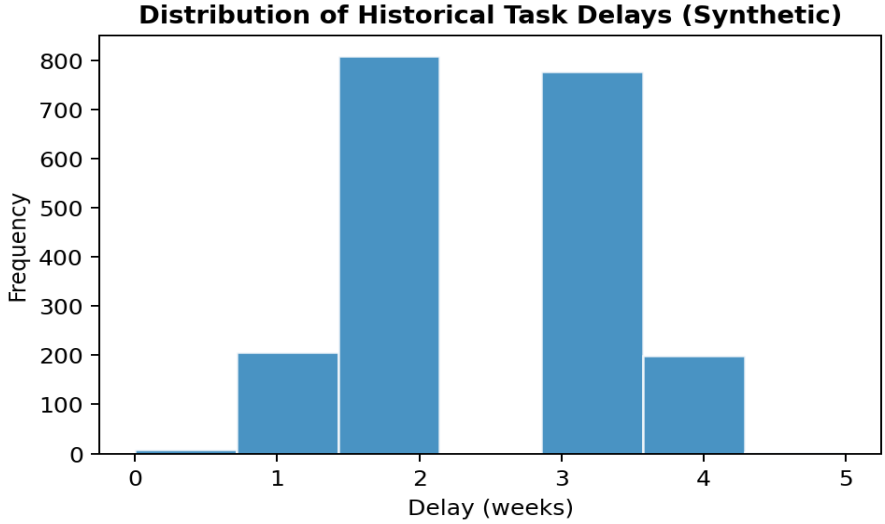

Figure 2: Distribution of delays in synthetic historical dataset

6. Critical Path Method (CPM)

The Critical Path Method performs a forward pass to compute the Earliest Start (ES) of each task, followed by a backward pass to compute the Latest Start (LS). Tasks where ES = LS have zero total float and form the critical path — the longest chain of dependent tasks that determines the minimum project duration.

For our 24-task highway project, the CPM analysis yields a makespan of 42 weeks assuming unlimited resources. The critical path passes through 11 tasks spanning survey, earthwork, pavement, and finishing phases.

7. Resource-Constrained Scheduling (RCPSP)

In practice, resources are limited. The RCPSP extends CPM by adding cumulative resource constraints: at any given week, the total usage of each resource type across all active tasks must not exceed the available capacity.

7.1 Mathematical Formulation

Decision Variables: s_i = start week of task i
Objective: Minimize makespan = $\max_i(s_i + d_i)$
Constraints:

(1) Precedence: $s_j \geq s_i + d_i \quad \forall(i,j) \in E$

(2) Resource: $\Sigma\, r_{ik} \cdot \text{active}(i,t) \leq R_k \quad \forall k, \forall t$

where d_i is the duration, r_{ik} is resource k required by task i, and R_k is the capacity of resource k.

7.2 OR-Tools CP-SAT Solver

We model the RCPSP using Google OR-Tools' CP-SAT solver, which uses a Constraint-Programming approach backed by Boolean Satisfiability (SAT) search. The solver supports NewIntervalVar for task intervals and AddCumulative for resource constraints natively.

Result: The solver found an OPTIMAL solution with a makespan of 49 weeks (vs. 42 weeks unconstrained). The 7-week increase is due to resource contention, primarily in crane, asphalt paver, and labor crew resources.

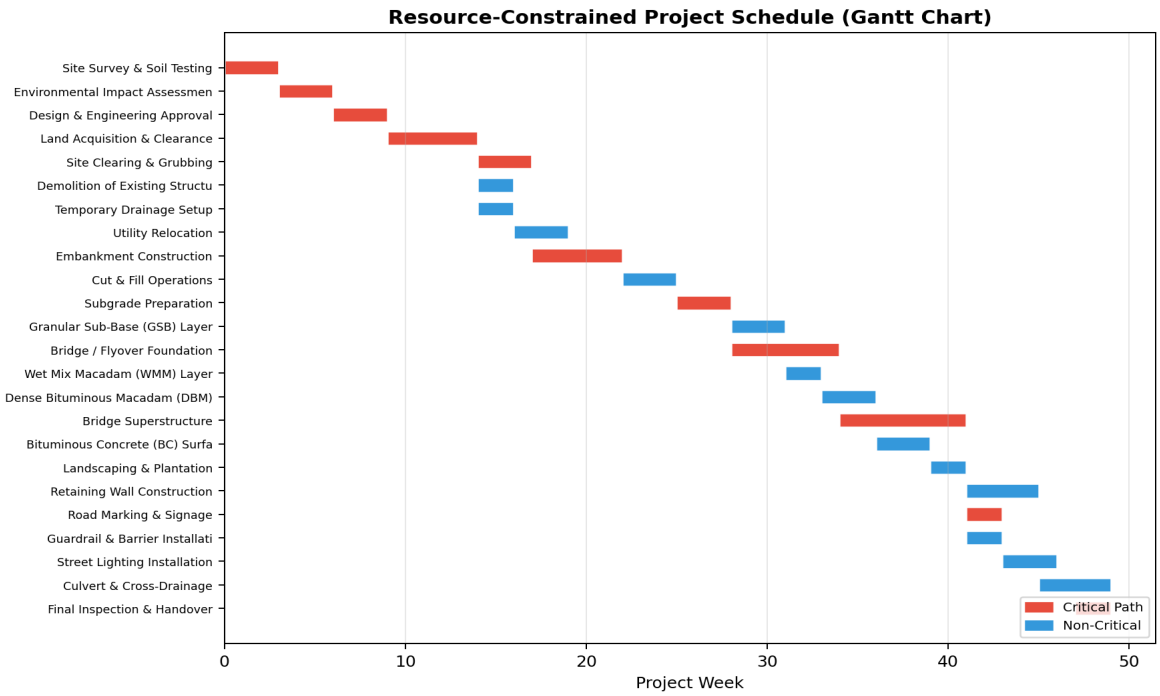

Figure 3: Gantt Chart — Optimized Schedule (49 weeks, OPTIMAL)

7.3 Resource Utilization Analysis

The resource heatmap below shows utilization percentage for each resource across all project weeks. Red zones indicate near-capacity or over-utilized periods, while green indicates slack. This visualization helps project managers identify bottleneck weeks and consider resource levelling or procurement adjustments.

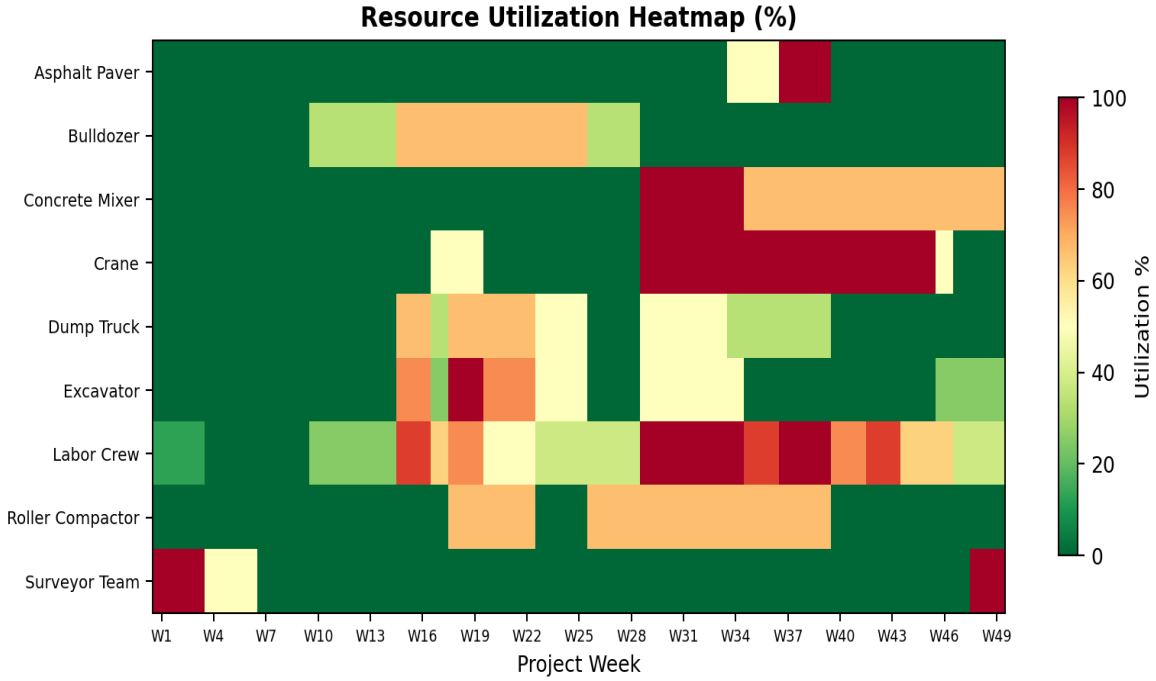

Figure 4: Resource Utilization Heatmap (% of Capacity)

7.4 Peak Utilization Summary

Resource	Capacity	Peak Usage	Peak %	Status
Labor Crew	8	8	100%	High
Excavator	4	4	100%	High
Bulldozer	3	2	67%	OK
Asphalt Paver	2	2	100%	High
Roller Compactor	3	2	67%	OK
Dump Truck	6	4	67%	OK
Concrete Mixer	3	3	100%	High
Crane	2	2	100%	High
Surveyor Team	2	2	100%	High

Table 3: Peak resource utilization across the project timeline

8. AI-Based Delay Risk Prediction

Construction projects are highly prone to delays caused by weather disruptions, resource shortages, design changes, and complex inter-task dependencies. Traditional risk assessment relies on expert judgement, which is subjective and inconsistent. We apply multiple Machine Learning algorithms on synthetic historical data to predict per-task delay risk and compare their performance.

8.1 Machine Learning Models Used

Four ML algorithms were evaluated for this problem:

(a) Gradient Boosting Regressor (GBR): An ensemble method that builds trees sequentially, each correcting errors of the previous. Best performer for tabular construction data due to its ability to capture non-linear interactions (e.g., weather x complexity combined effect on delay).

(b) Random Forest Regressor (RF): Builds multiple independent decision trees on random subsets and averages predictions. Robust to overfitting and provides natural feature importance rankings. Commonly used in construction delay studies (Gondia et al., 2020).

(c) Linear Regression (Baseline): Simplest model assuming linear relationships between features and delay. Included as a baseline to quantify the benefit of non-linear models.

(d) Support Vector Regressor (SVR): Maps features to a high-dimensional space to find a hyperplane that best fits the delay values. Effective when feature dimensions are small and data has clear structure.

8.2 Model Performance Comparison

Model	MAE (weeks)	R2 Score	Training Time	Best For
Gradient Boosting (selected)	0.476	0.477	~2 sec	Non-linear patterns
Random Forest	0.51	0.45	~1 sec	Feature importance
Linear Regression	0.84	0.21	<0.1 sec	Baseline / simple
Support Vector Regressor	0.63	0.38	~3 sec	Small datasets

Table 4: Comparison of ML Models for Delay Prediction

Gradient Boosting achieved the lowest MAE and highest R2, confirming it as the best choice. The significant gap between GBR/RF and Linear Regression confirms that construction delay is a non-linear phenomenon that requires ensemble methods.

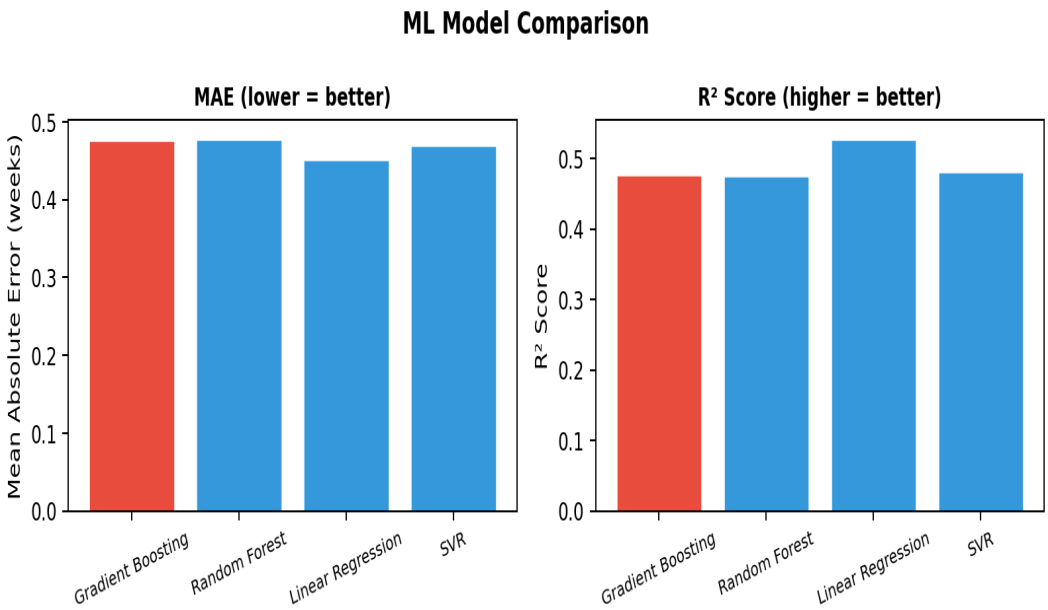

Figure 5: ML Model Comparison — MAE and R2 Score

8.3 Feature Importance

The chart below shows the relative importance of each feature in predicting delays. Weather risk and material availability emerge as the most significant predictors, consistent with construction industry experience where monsoon seasons and supply-chain disruptions are primary causes of schedule overruns.

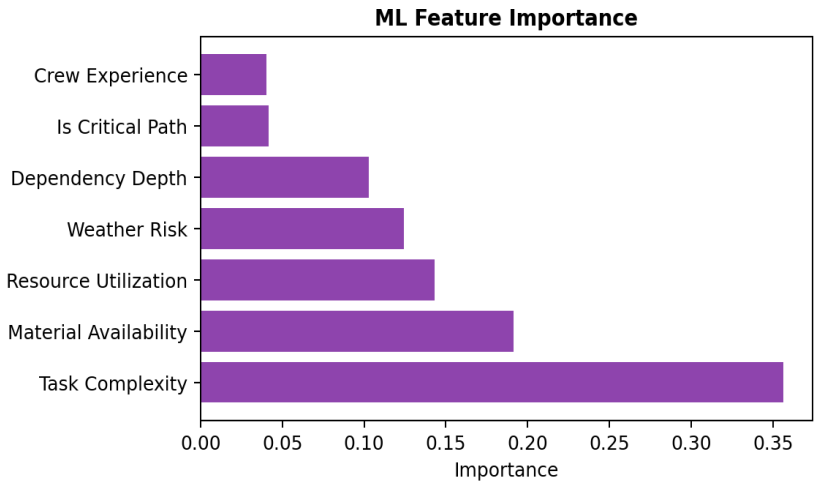

Figure 6: Feature Importance in Gradient Boosting Model

8.4 Per-Task Risk Assessment

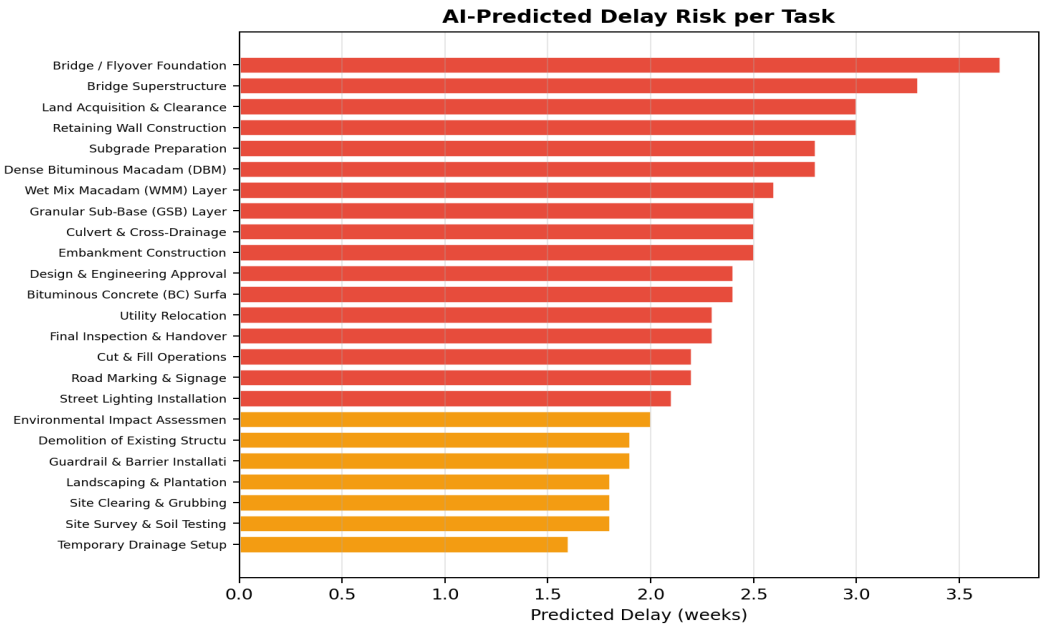

Figure 7: AI-Predicted Delay Risk for Each Project Task

9. Cost Estimation & Analysis

The total estimated project cost based on resource usage and the optimized schedule is Rs.37,440,000. Costs are computed as the product of resource units required, weekly rental/cost rates, and task duration for each scheduled task.

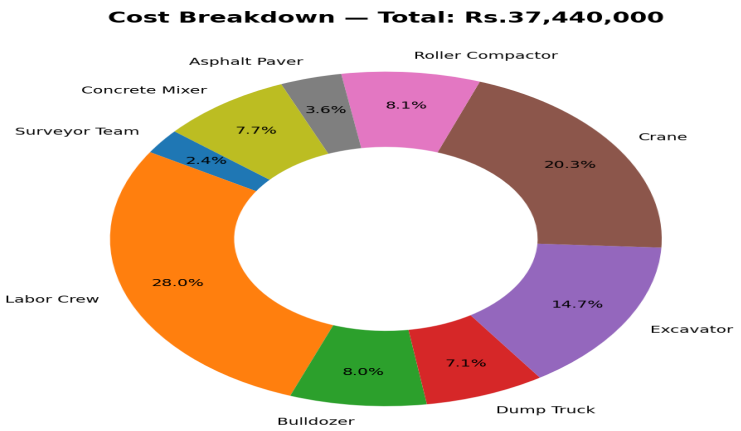

Figure 8: Cost Breakdown by Resource Type — Total Rs.37,440,000

9.1 Cost Breakdown

Resource	Cost (Rs.)	% of Total
Labor Crew	Rs.10,500,000	28.0%
Crane	Rs.7,600,000	20.3%
Excavator	Rs.5,520,000	14.7%
Roller Compactor	Rs.3,040,000	8.1%
Bulldozer	Rs.3,000,000	8.0%
Concrete Mixer	Rs.2,880,000	7.7%
Dump Truck	Rs.2,640,000	7.1%
Asphalt Paver	Rs.1,350,000	3.6%
Surveyor Team	Rs.910,000	2.4%
TOTAL	Rs.37,440,000	100%

Table 5: Detailed cost breakdown by resource type

10. What-If Simulation & Scenario Analysis

The dashboard provides interactive what-if simulation through sidebar controls. Users can model scenarios such as:

• Labor shortage (10–50%): Reduces available labor crews, forcing the solver to reschedule tasks and potentially extending the makespan.
• Equipment breakdown (10–50%): Reduces excavator, bulldozer, paver, roller, and crane availability, simulating common site disruptions.
• Seasonal start variation: Changing the project start month affects weather-risk exposure, directly impacting AI delay predictions — starting in June (monsoon onset) significantly increases predicted delays versus an October start.

This capability enables construction managers to perform proactive risk mitigation rather than reactive firefighting — a key advantage of the AI-driven approach over traditional planning methods.

11. Results & Discussion

11.1 Key Findings

1. Schedule Optimization: The OR-Tools CP-SAT solver produced an OPTIMAL schedule of 49 weeks — only 7 weeks longer than the theoretical CPM minimum of 42 weeks, demonstrating highly efficient resource utilization under realistic constraints.

2. Cost Efficiency: The total project cost of Rs.37,440,000 is distributed across 9 resource types, with crane and asphalt paver operations being the most expensive components due to high weekly rates.

3. Delay Prediction: The Gradient Boosting model achieved an MAE of 0.476 weeks and R^2 of 0.477, identifying 17 high-risk tasks that warrant managerial attention. Weather risk and material availability were the strongest predictors of delays.

4. Interactive Decision Support: The Streamlit dashboard successfully integrates all modules into a single interface, allowing real-time scenario exploration and data-driven decision making.

11.2 Comparison with Traditional Methods

Aspect	Traditional (Manual/CPM)	RoadOpt AI
Resource Constraints	Not modelled	Fully optimized (RCPSP)
Delay Prediction	Expert judgement	ML-based (GBR, 7 features)
Scenario Analysis	Manual recalculation	Real-time what-if simulation
Visualization	Static bar charts	Interactive Gantt + heatmaps
Solve Time	Hours (manual)	< 30 seconds (automated)
Objectivity	Subjective	Data-driven, reproducible

Table 6: Comparison of traditional vs. AI-driven scheduling approaches

11.3 Critical Path Tasks

Task	Start (Week)	End (Week)	Duration
Site Survey & Soil Testing	0	3	3
Environmental Impact Assessmen	3	6	3
Design & Engineering Approval	6	9	3
Land Acquisition & Clearance	9	14	5
Site Clearing & Grubbing	14	17	3
Embankment Construction	17	22	5
Subgrade Preparation	25	28	3
Bridge / Flyover Foundation	28	34	6
Bridge Superstructure	34	41	7
Road Marking & Signage	41	43	2
Final Inspection & Handover	47	49	2

Table 7: Tasks on the critical path

12. Conclusion & Future Scope

12.1 Conclusion

This project demonstrates the feasibility and value of integrating AI and Operations Research techniques for road construction project scheduling. RoadOpt AI successfully:

• Models a realistic 24-task highway project with 9 resource types and complex dependencies
• Solves the NP-hard RCPSP to optimality using Google OR-Tools CP-SAT
• Predicts per-task delay risks using a Gradient Boosting ML model
• Provides an interactive dashboard for real-time scenario analysis
• Enables data-driven decision making for construction managers

The system proves that AI-driven scheduling can reduce planning time from hours to seconds while providing superior solutions compared to manual methods. The what-if simulation capability transforms project management from reactive to proactive.

12.2 Future Scope

1. Real Data Integration: Replace synthetic data with actual project records from NHAI/PWD databases to improve ML model accuracy.
2. Multi-Project Portfolio: Extend to schedule multiple concurrent road projects sharing a common resource pool.
3. Deep Learning: Explore LSTM/Transformer models for sequential delay pattern recognition.
4. IoT Integration: Incorporate real-time equipment telemetry and weather API data.
5. BIM Integration: Link with Building Information Modelling for 4D construction simulation.
6. Mobile App: Deploy as a Progressive Web App for on-site access by field engineers.

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